Object Counting Survey Across Modalities

Final 3-Day Codex Computer-Use Guide for Collecting 500+ Candidate Papers

Prepared for Joana Owusu | Focus: images, videos, open-vocabulary/foundation-model counting, and emerging modalities such as 3D, remote sensing, microscopy, thermal, and event-camera counting.

Purpose. This document is a practical operating guide for using **Codex computer use + Codex coding workflows** to collect, clean, group, and prepare more than 500 candidate papers for a journal-level survey on object counting across modalities. It combines the large-corpus target from the earlier 500+ pipeline with the stronger API strategy from the collection-plan document, but rewrites everything as a Codex-first workflow.

Important boundary. Do not use Codex to scrape Google Scholar at scale or repeatedly automate Scholar searches. Use Google Scholar only as a supervised browser-based seed discovery tool. Use Codex for file creation, spreadsheet management, API-based expansion, deduplication, grouping, quality checks, BibTeX generation, and survey-outline creation.

1. Final Goal and Deliverables

The goal is not to read 500 papers in three days. The goal is to build a defensible survey corpus: a large candidate library, a screened literature matrix, a core reading set, and a documented search process that can be described in a journal paper.

Output	Target	Purpose
seed_papers_google_scholar.xlsx	150-250 high-quality seeds	Manual Scholar seed set collected with Codex computer-use support.
master_papers_raw.csv	900-2,000 raw records	All records from APIs, seeds, references, citations, query bank, and optional exports.
literature_matrix_clean.xlsx	500-700 deduplicated candidates	Main working corpus for the survey.
core_reading_set.xlsx	80-150 A-tier papers	Deep reading set for writing the survey.
references_validated.bib	BibTeX for all included papers	LaTeX-ready references with unique keys and escaped characters.
screening_log.xlsx	All keep/remove decisions	Transparent search and inclusion/exclusion record.
taxonomy_groups.md	Grouped literature map	Papers grouped by modality, method family, dataset, metrics, input/output type, and application.
survey_outline_final.md	Journal-ready outline	Survey structure with suggested tables, figures, and key papers.

2. How Codex Should Be Used

Codex mode	Use it for	Do not use it for
Computer/browser use	Supervised Scholar browsing, opening paper pages, copying seed metadata, checking PDFs, exporting citation files manually.	Unattended Scholar scraping, automated repetitive searches, bypassing CAPTCHA, or collecting pages at scale.
Local coding / terminal	Creating scripts, reading spreadsheets, querying APIs, deduplicating records, generating Excel/BibTeX/Markdown outputs.	Changing data without logs or deleting excluded papers without reasons.
Project/repo workflow	Maintaining a reproducible folder with data, scripts, logs, outputs, and README instructions.	One-off scripts with no saved query logs.

Codex mode	Use it for	Do not use it for
Chat assistance	Screening batches of abstracts, improving taxonomy labels, checking inclusion criteria, generating outline sections.	Blindly accepting automatic labels for A-tier papers.

3. Three-Day Timeline

Day	Main goal	End-of-day target
Day 1	Scholar seed collection + project setup	150-250 seed papers, exact queries logged, folder/repo ready, API keys configured.
Day 2	Codex API expansion to 500+ papers	900-2,000 raw records; 500-700 deduplicated and scored candidates.
Day 3	Screening, gap-filling, BibTeX validation, and survey organization	Core set, taxonomy groups, screening summary, validated BibTeX, and final outline.

4. Day 0 / Day 1 Morning Setup

4.1 Create the project folder

```
object_counting_survey_codex/
README.md
data/
seed_papers_google_scholar.xlsx
query_log.xlsx
raw_api_results/
master_papers_raw.csv
literature_matrix_clean.xlsx
screening_log.xlsx
scripts/
00_setup_project.py
01_collect_paperswithcode.py
02_collect_semantic_scholar.py
03_collect_openalex.py
04_collect_crossref.py
05_collect_arxiv.py
06_merge_deduplicate.py
07_score_group_screen.py
08_export_bibtex.py
09_generate_reports.py
outputs/
core_reading_set.xlsx
references.bib
references_validated.bib
taxonomy_groups.md
grouped_papers_final.md
benchmark_compendium.md
collection_summary.md
survey_outline_final.md
papers_pdf/
image/
video/
other_modalities/
datasets/
```

4.2 Install packages and set API variables

```
# Create and activate an environment if desired
python -m venv .venv
source .venv/bin/activate

# Install Python packages
pip install pandas openpyxl requests tqdm rapidfuzz bibtexparser pyyaml scikit-
learn matplotlib
```

```
# Optional but strongly recommended: Semantic Scholar API key
export S2_API_KEY="your_semantic_scholar_key_here"

# Use your email for OpenAlex polite-pool requests
export OPENALEX_MAILTO="your_email@domain.com"
```

4.3 Seed spreadsheet columns

Create **seed_papers_google_scholar.xlsx**. These columns are designed to support expansion, deduplication, grouping, and screening.

Column	Use
seed_id	Unique ID such as S001, S002.
paper_title	Full paper title.
authors	Main authors or full author list.
year	Publication year.
venue	CVPR, ICCV, ECCV, NeurIPS, WACV, arXiv, journal, etc.
source_found	Google Scholar, paper reference list, Semantic Scholar, Papers With Code, etc.
google_scholar_query	Exact query used to find the seed paper.
modality	Image; Video; Multimodal; 3D / Point Cloud; Remote Sensing / Aerial; Medical / Microscopy; Thermal; Event Camera; Other.
task_category	Survey; Dataset/Benchmark; Density-based; Detection-based; Few-shot; Class-agnostic; Zero-shot; Open-vocabulary; Foundation-model; Video Tracking-based; Application-specific.
input_type	Image; Video; Text Prompt; Exemplar Boxes; Exemplar Patches; Points; Masks; Depth/RGB-D; Point Cloud; Natural Language Query.
output_type	Count Only; Density Map + Count; Boxes + Count; Masks + Count; Tracks + Unique Count; Points + Count; Class-wise Counts; Benchmark.
dataset_or_benchmark	FSC-147, FSCD-147, CountBench, OmniCount, CARPK, ShanghaiTech, TAO-Count, VideoCount, etc.
main_contribution	One-sentence reason this paper matters.
relevance_score	High, Medium, Low at seed stage. Later converted to 0-5.
doi	DOI if available.
arxiv_id	arXiv ID if available.
paper_url	Landing page, arXiv, publisher, or PDF URL.
bibtex_collected	Yes/No.
pdf_saved	Yes/No.
notes	Any observation such as highly cited, dataset paper, recent, weak relevance.

5. Day 1: Google Scholar Seed Collection with Codex Computer Use

Day 1 is a supervised discovery step. Codex can help keep the spreadsheet clean and can guide browsing, but the work should stay manual and controlled. The target is 150-250 strong seeds, not thousands of Scholar results.

5.1 Codex computer-use prompt for Scholar seed collection

```
You are helping me collect seed papers for a journal survey on object counting
across modalities.
Use browser/computer actions only under my supervision. Do not scrape Google
```

```
Scholar.

Task:
1. Open my seed spreadsheet: data/seed_papers_google_scholar.xlsx.
2. For each Google Scholar query I give you, help me inspect only:
- top 20 relevance-sorted results
- top 10 date-sorted results
3. For each paper that is clearly relevant, fill the spreadsheet columns:
seed_id, paper_title, authors, year, venue, source_found, google_scholar_query,
modality, task_category, input_type, output_type, dataset_or_benchmark,
main_contribution, relevance_score, doi, arxiv_id, paper_url,
bibtex_collected, pdf_saved, notes.
4. Use the inclusion/exclusion rules in README.md.
5. Keep a query log with query, date searched, number inspected, number saved.
6. Stop after the target seed count is reached or after the allocated time.

Important:
- Do not automate repetitive Scholar requests.
- Do not bypass CAPTCHA or rate limits.
- If Scholar blocks or slows down, stop and switch to API sources or manual
exports.
```

5.2 Modality-first Scholar query plan

Bucket	Queries	Seed target
Survey / taxonomy	"object counting" "survey"; "visual counting" survey; "class-agnostic counting" survey; "few-shot object counting" survey; "open-vocabulary object counting" survey	15-25
Image counting - general	"object counting" "computer vision"; "visual object counting"; "dense object counting"; "object counting and detection"	25-40
Few-shot / exemplar-based	"few-shot object counting"; "exemplar-based object counting"; "class-agnostic object counting"; "FSC-147" "object counting"	40-60
Zero-shot / open-vocabulary	"zero-shot object counting"; "open-vocabulary object counting"; "text-guided object counting"; "vision-language" "object counting"	35-50
Foundation models	"object counting" "foundation models"; "object counting" "SAM"; "object counting" "CLIP"; "visual counting" "vision-language model"	20-35
Video counting	"video object counting"; "object counting in videos"; "open-world object counting in videos"; "unique instance counting" video; "CountVid"; "VideoCount"	35-50
Other modalities	"3D object counting"; "object counting" "point cloud"; "RGB-D object counting"; "event camera" object counting; "thermal" object counting	20-35
Domain modalities	"remote sensing" "object counting"; "aerial image" "object counting"; "cell counting" "computer vision"; "microscopy" "object counting"; "UAV" object counting	40-60
Datasets / benchmarks	"FSC-147"; "FSCD-147"; "CountBench" object counting; "OmniCount"; "CARPK" counting; "ShanghaiTech" counting; "TAO-Count"	30-50

5.3 Day 1 inclusion/exclusion rules

Decision	Rule
Include	Paper introduces a counting method, dataset, benchmark, evaluation protocol, survey taxonomy, or strong application case.
Include	Paper covers image, video, 3D, point cloud, RGB-D, aerial/remote sensing, microscopy, thermal, event-camera, or multimodal visual counting.
Include	Paper is highly cited, very recent, dataset-defining, method-defining, or directly related to few-shot, zero-shot, open-vocabulary, class-agnostic, foundation-model, or video counting.
Exclude	Paper only mentions counting in passing but is mainly generic object detection, segmentation, or tracking.

Decision	Rule
Exclude	Paper is unrelated to visual objects, images, videos, or visual modalities.
Exclude	Paper has no accessible metadata and no clear relevance to the survey taxonomy.

6. Taxonomy Rules for Categorization

6.1 When to select Multimodal

Select **Multimodal / Vision-Language** when the paper combines visual input with another signal, especially language. Examples include image/video + text prompt, VQA-style counting questions, CLIP/VLM/LLaVA-style counting, open-vocabulary counting from text labels, text + exemplars, RGB-D, image + point cloud, or image/video + sensor metadata.

```
Examples:
Image; Multimodal; Vision-Language -> image + text prompt counting
Video; Multimodal; Open-Vocabulary -> video + text prompt unique counting
3D; RGB-D; Multimodal -> RGB image + depth map
Image -> image-only exemplar counting
Video -> video-only tracking/counting without
text prompt
```

6.2 Input type and output type rules

Field	Meaning	Examples
input_type	What the method needs before counting.	Image; Video; Text Prompt; Exemplar Boxes; Exemplar Patches; Points; Masks; RGB-D / Depth; Point Cloud; Remote Sensing Image; Medical/Microscopy Image; Natural Language Query.
output_type	What the method returns after counting.	Count Only; Density Map + Count; Bounding Boxes + Count; Masks + Count; Points + Count; Tracks + Unique Count; Class-wise Counts; Region-wise Counts; Question Answer.

6.3 Example labels

Paper type	Modality	Input Type	Output Type
Traditional crowd counting	Image	Image	Density Map + Count
Few-shot FSC-147 method	Image	Image; Exemplar Boxes/Patches	Count Only
FSCD-147 detection-counting	Image; possibly Multimodal	Image; Exemplar Boxes or Text Prompt	Boxes + Count
Zero-shot text-guided counting	Image; Multimodal	Image; Text Prompt	Count Only or Boxes + Count
SAM/CLIP open-vocabulary counting	Image; Multimodal; Vision-Language	Image; Text Prompt	Masks + Count or Boxes + Count
Video unique counting	Video; possibly Multimodal	Video; Text Prompt or Exemplar	Tracks + Unique Count
3D point-cloud counting	3D / Point Cloud	Point Cloud or RGB-D	Count Only or 3D Boxes + Count

7. Day 2: Codex API Expansion to 500+ Papers

Day 2 is where Codex does the heavy work. The pipeline should combine seed-paper expansion with direct query-bank searches. Use Papers With Code, Semantic Scholar batch endpoints, OpenAlex, Crossref, arXiv, and optional PubMed/domain APIs. The aim is 900-2,000 raw records and 500-700 retained candidates after deduplication and relevance scoring.

7.1 API sources

Source	Use in pipeline
Papers With Code	Task/dataset pages, benchmark-related papers, leaderboards, dataset/method links where available.
Semantic Scholar	Batch metadata, abstracts, citation counts, references, citations, related paper expansion.
OpenAlex	Large scholarly metadata graph, venues, concepts, cited-by counts, open access links, DOI lookup.
Crossref	DOI validation, publisher metadata, BibTeX support.
arXiv	Recent preprints, cs.CV/cs.RO/q-bio searches, arXiv IDs, abstracts, BibTeX entries.
PubMed/domain APIs	Optional for microscopy, cell counting, biomedical imaging, and healthcare applications.

7.2 Codex master prompt: build and run the full pipeline

```
I am building a reproducible literature collection pipeline for a journal survey
titled:
"Object Counting Across Modalities: A Survey of Image, Video, Open-Vocabulary,
and Emerging Visual Counting Methods."

You are Codex working in my project folder: object_counting_survey_codex/.
Use local files, Python scripts, and legitimate scholarly APIs. Do not scrape Google
Scholar.

Input file:
- data/seed_papers_google_scholar.xlsx

The seed file columns are:
seed_id, paper_title, authors, year, venue, source_found, google_scholar_query,
modality, task_category, input_type, output_type, dataset_or_benchmark,
main_contribution, relevance_score, doi, arxiv_id, paper_url,
bibtex_collected, pdf_saved, notes.

Build a complete Python pipeline that:
1. Creates the folder structure if missing.
2. Loads the seed spreadsheet.
3. Builds a query_bank.yaml with modality-specific and task-specific queries.
4. Queries these sources:
- Papers With Code API
- Semantic Scholar Graph API, using batch endpoints whenever possible
- OpenAlex API with mailto from OPENALEX_MAILTO
- Crossref API for DOI and metadata resolution
- arXiv API for recent/gap-fill papers
5. Expands each seed using:
- references
- citing papers
- related papers where available
- query-bank keyword searches
- dataset/benchmark keyword searches
6. Saves all raw records to data/master_papers_raw.csv.
7. Deduplicates using DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID,
normalized title, and fuzzy title matching.
8. Assigns modality, method_family, input_type, output_type, application_area,
relevance_score, tier, and reason_kept_or_removed.
9. Saves outputs:
- data/literature_matrix_clean.xlsx
- data/screening_log.xlsx
- outputs/core_reading_set.xlsx
- outputs/references.bib
- outputs/taxonomy_groups.md
- outputs/collection_summary.md
- outputs/survey_outline_final.md
10. Logs every API query, date/time, number returned, number kept, and errors.

Before running long API calls, show me the planned scripts and ask for permission to
proceed.
```

7.3 query_bank.yaml content

```
survey_taxonomy:
- object counting survey
- visual counting survey
- class agnostic counting survey
- few-shot object counting survey
image_counting:
- object counting computer vision
- visual object counting
- dense object counting
- object counting and detection
few_shot_exemplar:
- few-shot object counting
- exemplar-based object counting
- class-agnostic object counting
- FSC-147 object counting
zero_shot_open_vocab:
- zero-shot object counting
- open-vocabulary object counting
- text-guided object counting
- vision-language model object counting
foundation_models:
- object counting foundation models
- SAM object counting
- CLIP object counting
- vision language model visual counting
video_counting:
- video object counting
- open-world object counting in videos
- unique instance counting video
- tracking based object counting video
- re-identification object counting video
other_modalities:
- 3D object counting
- point cloud object counting
- RGB-D object counting
- event camera object counting
- thermal object counting
applications:
- remote sensing object counting
- aerial image object counting
- UAV object counting
- microscopy cell counting
- traffic object counting
- wildlife object counting
- agriculture object counting
benchmarks:
- FSC-147 object counting
- FSCD-147 object counting
- CountBench object counting
- OmniCount object counting
- CARPK counting
- ShanghaiTech counting
- TAO-Count
- VideoCount
- MOT20-Count
```

7.4 Codex prompt: implement API scripts

```
Create the following Python scripts with clear functions, logging, retries, and
rate-limit handling:

scripts/01_collect_paperswithcode.py
- Query Papers With Code task and dataset endpoints for object counting terms.
- Extract title, authors, year, arxiv_id, url, dataset, metrics, source.
- Save data/raw_api_results/paperswithcode_results.csv.

scripts/02_collect_semantic_scholar.py
- Use S2_API_KEY if available.
- Use batch endpoints where possible.
- Resolve seed titles/DOIs.
- For each resolved seed, collect metadata, references, citations, and abstracts.
- Cap raw expansion to 2,000 records unless I change the limit.
- Save data/raw_api_results/semantic_scholar_results.csv.

scripts/03_collect_openalex.py
- Use OPENALEX_MAILTO.
```

```
- Query works by DOI and query-bank terms.
- Reconstruct abstracts from inverted indexes.
- Save data/raw_api_results/openalex_results.csv.

scripts/04_collect_crossref.py
- Resolve missing DOI values using title + first author.
- Save data/raw_api_results/crossref_resolved.csv.

scripts/05_collect_arxiv.py
- Search arXiv for query-bank terms and gap-fill terms.
- Save data/raw_api_results/arxiv_results.csv.

Each script must be safe to rerun and must not duplicate records in its output.
```

7.5 Codex prompt: deduplicate, score, and group

```
Read all files in data/raw_api_results/ plus data/seed_papers_google_scholar.xlsx.
Create data/master_papers_raw.csv and data/literature_matrix_clean.xlsx.

Deduplicate using:
- DOI exact match
- arXiv ID exact match
- Semantic Scholar paper ID exact match
- OpenAlex work ID exact match
- normalized title exact match
- fuzzy title similarity >= 92%

For each remaining paper, assign:
- modality: image, video, multimodal, 3D, point_cloud, RGB-D, remote_sensing,
microscopy, thermal, event_camera, medical, other
- method_family: density_map, regression_based, detection_based, exemplar_based,
few_shot, class_agnostic, zero_shot, open_vocabulary, vision_language,
segmentation_based, tracking_based, re_identification, foundation_model,
dataset_benchmark, survey, application_specific
- input_type: image, video, text_prompt, natural_language_query, exemplar_boxes,
exemplar_patches, points, masks, depth, RGB-D, point_cloud,
remote_sensing_image,
microscopy_image, multimodal
- output_type: count_only, density_map_plus_count, boxes_plus_count,
masks_plus_count, points_plus_count, tracks_plus_unique_count,
classwise_counts, regionwise_counts, question_answer, benchmark
- application_area: general, agriculture, traffic, wildlife, healthcare,
microscopy,
remote_sensing, manufacturing, crowd_counting, autonomous_driving, other
- relevance_score: integer 0-5
- tier: A_core, B_important, C_background, Exclude
- reason_kept_or_removed: one sentence

Scoring:
5 = central method/dataset/survey for object counting
4 = directly relevant to one modality or task family
3 = useful background or application paper
2 = weakly related, keep only if needed for breadth
1 = mostly unrelated
0 = remove

Targets:
- Retain 500-700 candidate papers in literature_matrix_clean.xlsx.
- Place 80-150 papers in A_core.
- Put excluded papers in a separate sheet with reasons.
- Print counts by source, modality, method_family, year, and tier.
```

8. Day 3: Screening, Gap Filling, and Journal-Ready Organization

Day 3 turns the large corpus into a writing-ready structure. The final matrix should not simply list papers; it should prove coverage, show trends, reveal gaps, and support a clear journal survey argument.

8.1 Coverage targets

Area	Minimum target in clean matrix	Gap-fill query if below target
Image counting	150+	few-shot object counting; class-agnostic counting; dense object counting
Zero-shot/open-vocabulary/foundation models	60+	zero-shot object counting; open-vocabulary object counting; SAM object counting
Video counting	40+	video object counting; unique instance counting; CountVid; VideoCount; TAO-Count
3D / point cloud / RGB-D	20+	3D object counting; point cloud object counting; RGB-D object counting
Remote sensing / aerial / UAV	35+	remote sensing object counting; aerial vehicle counting; UAV object counting
Microscopy / medical / cell	35+	cell counting computer vision; microscopy object counting; medical image counting
Datasets / benchmarks / surveys	50+	FSC-147; FSCD-147; CountBench; OmniCount; TAO-Count; object counting survey

8.2 Codex prompt: gap-fill missing modalities

```
Read data/literature_matrix_clean.xlsx and count papers by modality,
method_family,
and dataset_or_benchmark.

If any target is below the minimum, run arXiv, Semantic Scholar, and OpenAlex gap-
fill
queries for that category. Add only papers not already in the matrix.

Gap-fill categories:
- video object counting / unique instance counting / CountVid / VideoCount / TAO-
Count
- 3D object counting / point cloud counting / RGB-D counting
- remote sensing object counting / aerial object counting / UAV counting
- microscopy cell counting / medical image counting
- open-vocabulary object counting / zero-shot object counting / foundation model
counting

For each added paper:
- Fill all taxonomy columns.
- Add reason_kept_or_removed.
- Update references.bib.
- Update data/screening_log.xlsx.

Print: number of new papers added per category and final coverage counts.
```

8.3 Codex prompt: manual screening support

```
I want to manually review uncertain papers.
Open data/literature_matrix_clean.xlsx and create a new file:
outputs/manual_review_batch.xlsx

Select papers where:
- relevance_score is 2 or 3, OR
- tier is C_background, OR
- modality/method_family is unknown, OR
- abstract is missing but title looks relevant.

For each paper, create columns:
review_id, title, year, venue, abstract, current_modality, current_method_family,
recommended_decision, reason, reviewer_notes.

Also create a Markdown file outputs/manual_review_questions.md with 20-paper
batches.
For each batch, include title, abstract, and the question:
KEEP, DOWNGRADE, or REMOVE? Give one sentence reason.
```

8.4 Codex prompt: final taxonomy and outline

```
Read the final data/literature_matrix_clean.xlsx.

Generate:
1. outputs/taxonomy_groups.md
For each group, list representative papers, foundational papers, recent papers,
datasets, metrics, limitations, and open gaps.

2. outputs/grouped_papers_final.md
Group papers by:
- modality
- method family
- input type
- output type
- dataset/benchmark
- application area

3. outputs/benchmark_compendium.md
Table columns:
Dataset | Modality | Task setting | Input prompt | Annotation type |
Output | Metrics | Common methods | Notes

4. outputs/survey_outline_final.md
Create a journal survey outline with:
- section title
- 2-sentence purpose
- 3-7 key papers to cite
- suggested table or figure
- open questions/gaps for that section

5. outputs/collection_summary.md
Include:
- total raw records
- duplicates removed
- retained papers
- counts by source, year, modality, method_family, dataset, and tier
- top 20 most-cited papers
- final A/B/C tier counts
- recommended reading order.
```

9. BibTeX and LaTeX Validation

```
Read outputs/references.bib and validate every entry.

Rules:
1. Escape LaTeX special characters in all fields:
& -> \&
% -> \%
$ -> \$
# -> \#
_ -> \_
2. Check balanced braces.
3. Check duplicate BibTeX keys.
4. Use key format:
FirstAuthorLastname_Year_FirstImportantTitleWord
5. Required fields:
@article: author, title, journal, year
@inproceedings: author, title, booktitle, year
@misc / arXiv: author, title, year, eprint, archivePrefix
6. Save outputs/references_validated.bib.
7. Create outputs/bibtex_validation_report.md with entries checked, fixed,
warnings,
and unresolved metadata.
```

10. PRISMA-Style Search and Screening Log

A journal survey is stronger when it can explain how the literature was collected. Codex should produce a simple PRISMA-style summary even if the final paper does not include a formal PRISMA diagram.

```
Create outputs/screening_summary.md and data/screening_log.xlsx.
Report:
```

```
- number of Google Scholar seed papers collected manually
- number of records retrieved from Papers With Code
- number of records retrieved from Semantic Scholar
- number of records retrieved from OpenAlex
- number of records retrieved from Crossref
- number of records retrieved from arXiv
- number of raw records before deduplication
- number removed as duplicates
- number screened by title/abstract
- number excluded and top exclusion reasons
- number retained in the final 500+ candidate corpus
- number in A_core, B_important, C_background, and Exclude tiers
- coverage counts by modality and method_family

Also create a simple text PRISMA flow that can be converted into a figure later.
```

11. Quality Control Rules for Journal Acceptance

Quality rule	Codex instruction
Reproducibility	Record every query, source, date searched, API endpoint, number returned, number kept, and reason for exclusion.
Coverage	Ensure coverage of images, videos, open-vocabulary/foundation models, and at least three emerging/specialized modalities.
Balance	Do not let crowd counting, generic detection, or a single dataset dominate the corpus unless explicitly scoped.
Recency	Include strong recent papers from 2022-2026, especially zero-shot, open-vocabulary, foundation-model, and video counting papers.
Foundations	Keep older foundational papers if they introduced a dataset, metric, or method family.
Taxonomy	Every included paper must map to at least one modality and one method family.
Manual review	Manually inspect all A-tier papers and uncertain B-tier papers.
No Scholar scraping	Scholar is for supervised seed discovery only; bulk expansion must use APIs and exports.

12. Copy-Ready Day-by-Day Operating Plan

Time	Task	Codex role
Day 1 morning	Create repo, environment, spreadsheet, query log, taxonomy README.	Generate folder structure, templates, validation checks.
Day 1 afternoon	Run controlled Scholar searches for surveys, image counting, few-shot, zero-shot, open-vocabulary, and foundation models.	Assist browser navigation and spreadsheet filling under supervision.
Day 1 evening	Run video, 3D, remote sensing, microscopy, thermal, event-camera, and benchmark searches.	Clean seed spreadsheet, fill missing columns, summarize seed counts.
Day 2 morning	Build API scripts and query_bank.yaml.	Write safe, rerunnable Python scripts with logging and rate limits.
Day 2 afternoon	Run API expansion.	Collect raw records from PWC, Semantic Scholar, OpenAlex, Crossref, arXiv.
Day 2 evening	Merge, deduplicate, score, and produce 500-700 candidate papers.	Generate literature_matrix_clean.xlsx and screening logs.
Day 3 morning	Check coverage and gap-fill weak categories.	Run gap-fill queries and update matrix.
Day 3 afternoon	Manual review of A-tier and uncertain B/C papers.	Prepare review batches and update decisions.

Time	Task	Codex role
Day 3 evening	Generate final BibTeX, taxonomy groups, benchmark compendium, collection summary, and survey outline.	Produce journal-ready outputs and validation reports.

13. Final Checklist Before Writing

- At least 500 deduplicated candidate papers are retained in data/literature_matrix_clean.xlsx.
- At least 80-150 A-tier papers are identified for deep reading.
- Every retained paper has modality, method_family, input_type, output_type, dataset_or_benchmark, relevance_score, tier, and reason_kept_or_removed.
- The corpus covers images, videos, multimodal/open-vocabulary counting, and at least three emerging modalities or application domains.
- references_validated.bib compiles without obvious duplicate keys or broken special characters.
- screening_log.xlsx explains why papers were kept or excluded.
- taxonomy_groups.md shows clear sections for images, videos, 3D/point-cloud/RGB-D, remote sensing, microscopy, thermal/event-camera, and multimodal counting.
- survey_outline_final.md is organized by taxonomy and research gaps, not just chronological listing.
- collection_summary.md contains counts by source, year, modality, method family, dataset, and tier.
- No automated scraping of Google Scholar was used.

14. Notes on Source Documents and Codex Assumptions

This guide was prepared by combining two prior planning documents: the large-corpus pipeline targeting 500+ organized papers and the collection plan emphasizing Papers With Code, Semantic Scholar batch endpoints, arXiv gap filling, and BibTeX validation. The guide keeps the 500+ paper target while rewriting the workflow specifically for Codex computer use and Codex local coding.

Codex should be treated as an agent that can help read, edit, and run code in a selected project directory, while browser/computer actions should remain supervised for Google Scholar and other websites. Keep the final process reproducible through saved scripts, logs, and spreadsheets.

Suggested first command to give Codex after creating the project folder:

```
Read README.md and data/seed_papers_google_scholar.xlsx.
Then inspect the project folder and create the missing scripts for the 3-day
literature
collection pipeline. Before running any long API collection, summarize your plan
and ask me
to approve the run.
```